GLYPH-BASED VISUALIZATION





Information Visualization - 4DV805

Glyph-based visualization

- Representing data through visual representations, which are called glyphs.
- Size, form, color, position, texture.
- Can be traced back in history to ancient times.
- Glyphs are a visual sign that differs from icons, since they are precoded in the computer and is a variant of a sign in the alphabet.
- Glyphs are a part of:
  - Typography
  - Coding
  - UI/UX



Glyphs in UI/UX

In UI glyphs can be used to create visual channels such as shape, colour, texture, size, orientation, aspect ratio or curvature, which creates depiction of multi-dimensional data attributes.

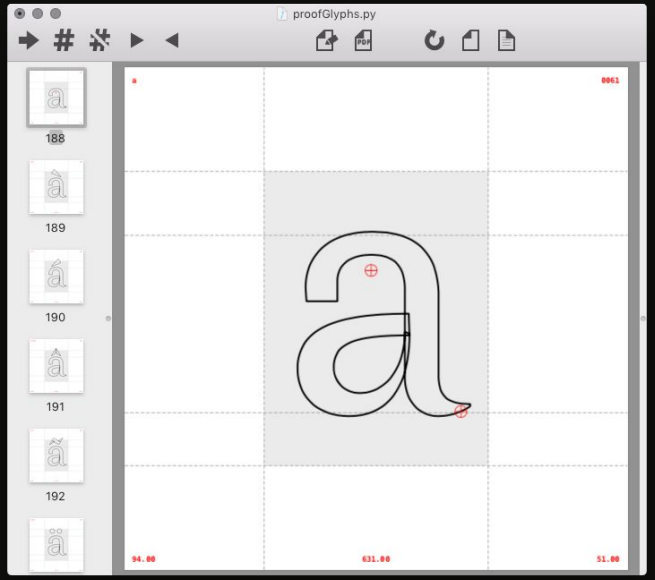
Glyphs are recognizable on their own but together they can create visual encoding.

Glyphs can create effective information encoding and visual communication

Glyphs also have the power to affect how information is perceived.

Cognitive overload vs guiding the user effectively.

Decreasing the learning curve.



Using to many glyphs and icons counteract their meaning of being a effective way of displaying information



Glyphs can be used to create:







Icons

Comprised of: images, metaphors and diagrams.

All display different qualities such as: Sensory qualities, relational and structural qualities, representative character of an object.

Index

Displays connection between a sign and its referent.

Symbol

Meaning is established through social and cultural constructions.

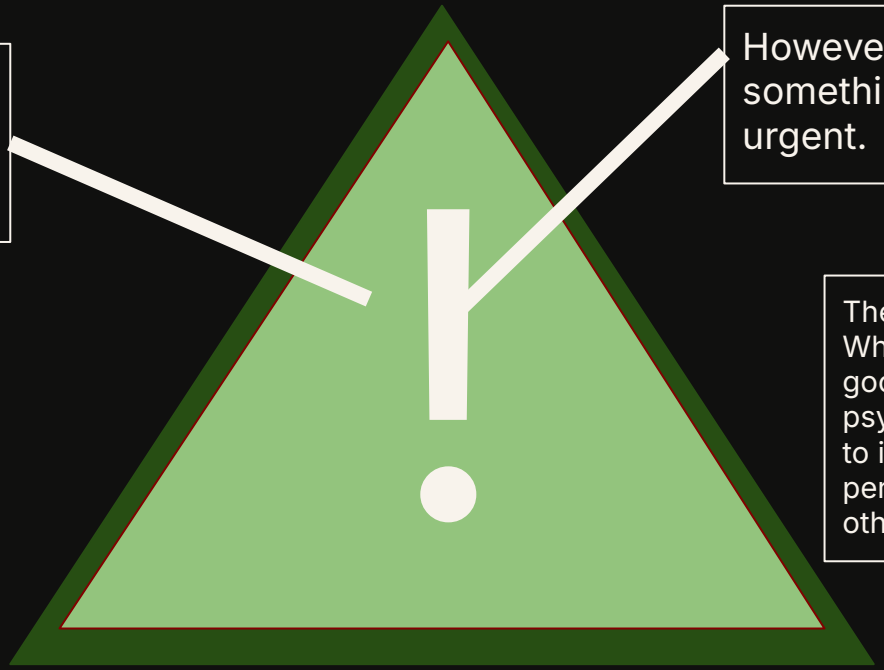
Meaning is created by us as people and the common world view.

What does this symbol tell you?



What does this symbol tell you?

The color tells you it’s something good or safe.



However the ! tells you it’s something bad, unsafe or urgent.

The symbol is contradicting. Whether it displays something good or bad depends on your psychology and how you choose to interpret it. Usually one perception of it will win over the other.

Variables affect the message

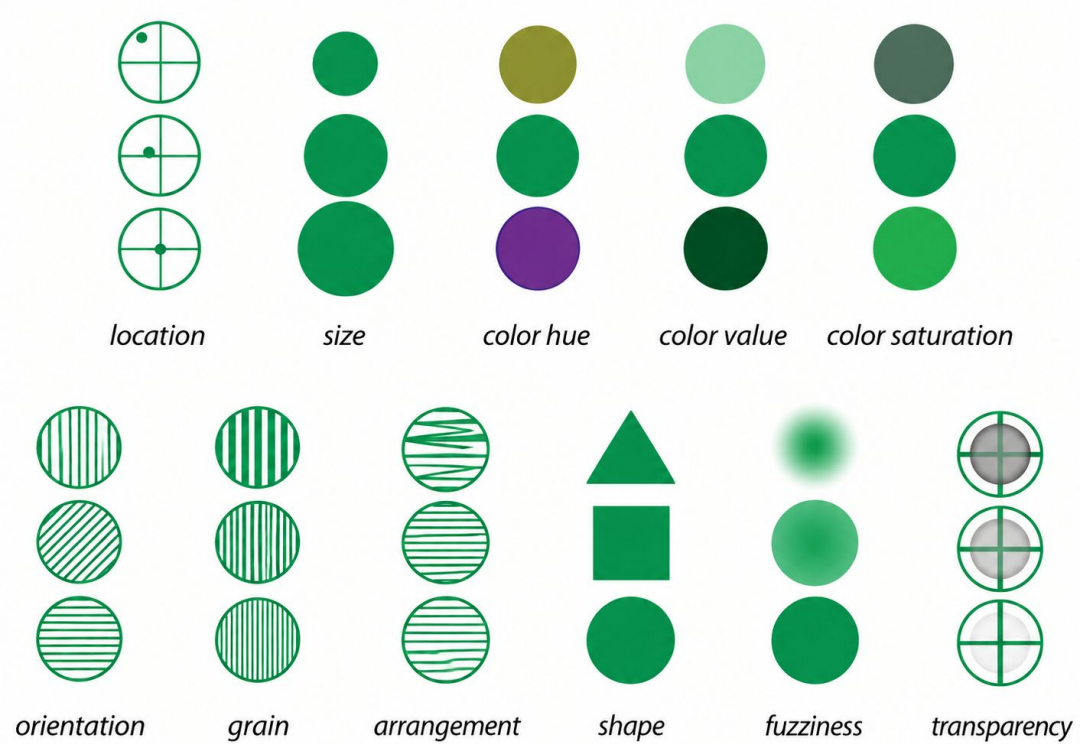
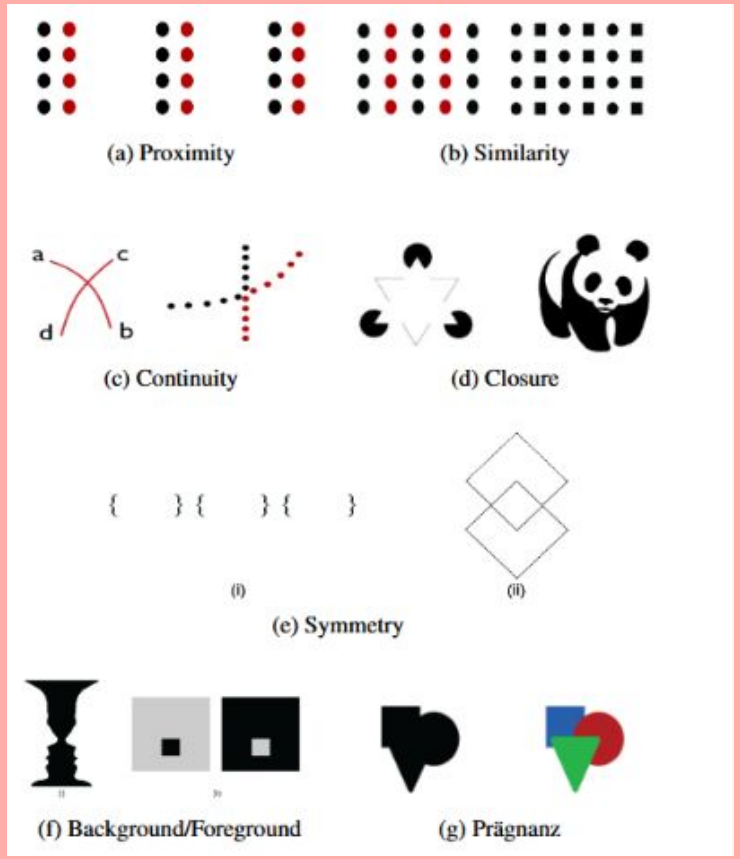


Figure 5: Visual Variables [Mac04].

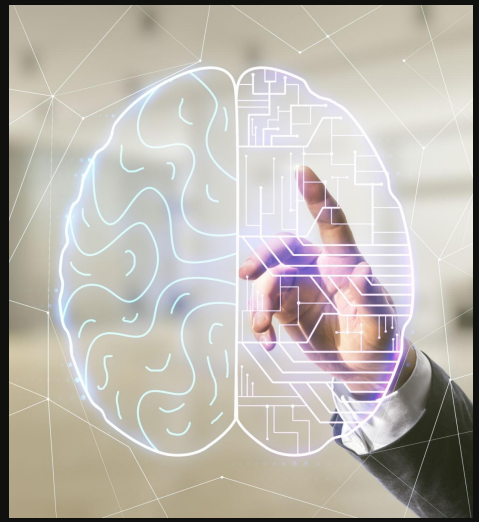
Gestalt principles

- Proximity
- Similarity
- Continuity
- Closure
- Symmetry
- Background/Foreground
- Prägnanz



Coding

- **Coding** is a fundamental concept in semiotics
- Creates a relationship between:
  - **Signifier** → what we see/hear
  - **Signified** → what it represents/means
- Interpretation depends on **prior knowledge**
- We can only recognise and interpret what we already **know or understand**



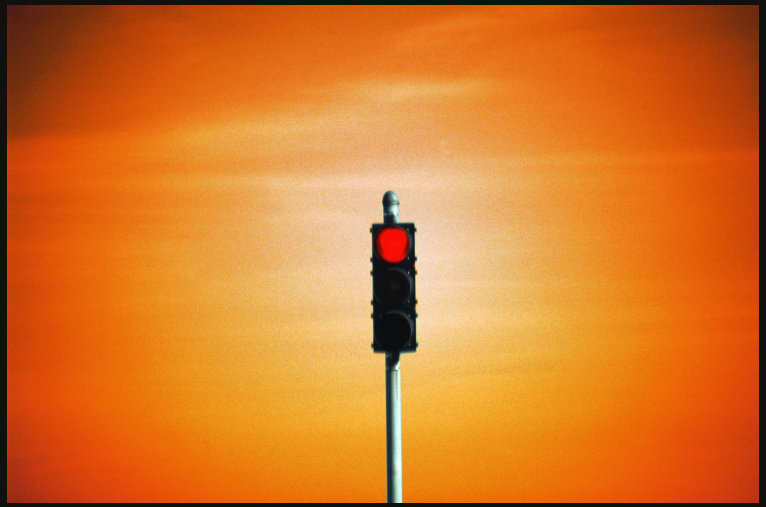
Example:

- **Social codes** – influenced by social and cultural conventions
- **Verbal language** - phonological
- **Bodily codes** - proximity, facial expression, gestures etc.

Diadic and triadic model

Diadic model – Ferdinand de Saussure

- A sign consists of two parts (signifier & signified)
- Example: Traffic light
  - **Signifier –** the red color
  - **Signified –** "STOP"



Triadic Model – Charles Sanders Peirce

- **three parts**:
  - **Representamen** – the sign itself
  - **Object** – what the sign refers to
  - **Interpretant** – the meaning interpreted by the viewer

Example: Cigarette

- Representamen → image/symbol of a cigarette
- Object → cigarette/smoking
- Interpretant → "smoking"

The **interpretation of the sign** is part of the model.

In interaction design

Signifiers

Visual clues



Affordances

The design tells you what you can do



Thank you!



